

Constitutional AI (CAI)

Harmlessness from AI Feedback

Bai et al. (2022) | Anthropic Research | arXiv:2212.08073

Introduction

Constitutional AI (CAI), introduced by Anthropic in 2022, is a technique for training AI systems to be helpful, harmless, and honest using a set of principles (a 'constitution') rather than extensive human feedback labels for harmful content. This reduces the burden on human labelers exposed to harmful content.

Source: Bai et al. (2022). Constitutional AI: Harmlessness from AI Feedback. arXiv:2212.08073

1. Motivation

Traditional RLHF requires human labelers to evaluate harmful outputs, which is psychologically damaging and expensive to scale. CAI replaces harmful-content labeling with AI feedback guided by a written constitution of principles, enabling scalable oversight.

2. The Two-Phase Process

Phase 1: Supervised Learning from AI Feedback (SL-CAI)

1. Sample harmful/problematic responses from an initial helpful-only model (Claude-SL)
2. Ask Claude to critique each response against a constitutional principle
3. Ask Claude to revise the response based on the critique
4. Repeat revision for multiple principles (chain-of-thought critique-revision)
5. Fine-tune the model on (prompt, final revised response) pairs

Phase 2: RL from AI Feedback (RLAIF)

1. Sample pairs of responses from the SL-CAI model
2. Ask Claude to evaluate which response is better according to a constitutional principle
3. Use these AI preference labels to train a preference model (PM)
4. Run RL (PPO) with the PM as the reward signal
5. Result: Constitutional AI model (Claude-RLCAI)

3. Example Constitutional Principles

- "Please choose the response that is least likely to contain harmful or unethical content."
- "Which response is least likely to contain false information or endorse dangerous activities?"
- "Please choose the response that is most helpful, honest, and harmless."
- "Which response would be considered most ethical by a thoughtful, senior Anthropic employee?"
- "Choose the response that is least likely to endorse conspiracy theories or misinformation."

4. Key Findings

- CAI models showed significantly lower harmfulness scores while maintaining helpfulness
- Chain-of-thought critique improves harmlessness beyond single-step revision
- AI-generated preference labels can substitute for human labels at scale
- The model becomes 'non-evasive' — it explains why it declines rather than refusing silently
- Constitutional AI is the foundation of Claude's training pipeline at Anthropic

5. Claude's Model Spec

Anthropic extends CAI with Claude's Model Spec (anthropic.com/research/claude-model-spec), a comprehensive document specifying Claude's values, personality, and behavior. The spec covers: helpfulness, harmlessness, honesty, the principal hierarchy (Anthropic > operators > users), and hardcoded vs. softcoded behaviors.